Universidad de Alcalá | Escuela Politécnica Superior

Operating Systems

780007EN · Group 2E

Universidad de Alcalá · EPS · Session S01 · 7 September 2026

Who am I?

Óscar García Población — English group coordinator

- Office hours: **Monday 10:00–12:00** and **Tuesday 12:00–14:00** (same schedule across ASO-GIT, SOG and SOA)
- Office: `DE-238`
- Spanish group coordinator: **David Fernández Barrero**

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Any Erasmus student in the audience today?

Course description

Operating Systems is a basic course of the first course of the Degree in Computer Engineering. It is the first course of the Operating Systems track, which continues with **Advanced Operating Systems** and, optionally, **Embedded Systems** and **Real-Time Systems**.

Operating systems provide abstraction levels high enough to build complex systems on top of hardware, in a simple and safe way — their evolution is closely tied to Computer Architecture.

Course platform

- **Blackboard** — https://uah.blackboard.com/
  - The forum is the **official communication channel**
  - The gradebook is kept in this platform
- **Discord** — course chat channel

Prerequisites

- **Computer Technology** (Tecnología de Computadores)
- **Basic Programming I** (Programación Básica I)
- You need to know how to program in **C**

Class schedule

- **Monday**
  - Laboratory: 12:00–13:55 · `EL4`
  - Theory: 15:00–16:55 · `NA8`
- From **7 September 2026** to **18 December 2026**

Syllabus

Theory contents

Operating Systems (this course)

P1 — Introduction to OS & Computer Architecture

Why do we need an OS at all? We start from bare hardware and see what gap system software must fill.

P2 — Operating Systems Structure

Now the why is clear, we look at the how: kernel, system calls, and the different ways to organize that software.

P3 — Processes and Threads

With the OS's shape in place, we meet its core abstraction: the running program, and the threads that share it.

P4 — Process Synchronization

When threads share data, things can go wrong — we learn to coordinate them safely.

P5 — CPU Scheduling

With many processes ready to run, the OS must decide who goes next, and for how long.

Advanced Operating Systems (next year)

P6 — Memory Management

Each scheduled process needs its own memory space — how the OS allocates and protects it.

P7 — Virtual Memory

Physical memory runs out; virtual memory makes each process believe it has more than there really is.

P8 — Input/Output

Processes need to talk to the outside world — how the OS mediates access to devices.

P9 — File Systems

Finally, persistent storage: how data outlives the process that created it.

Laboratory syllabus

- The Linux console
- The architecture simulator
- How to write programs
- How to use the Operating System services

Full schedule

Weeks 1–15

Full schedule (I)

Session	Date	Theory (`NA8`)	Lab (`EL4`)
`S01`	`7 Sep`	Course presentation + Introduction	—
`S02`	`14 Sep`	Introduction	The Linux Console (1/2)
`S03`	`21 Sep`	The structure of the OS	The Linux Console (2/2)
`S04`	`28 Sep`	The structure of the OS	Architecture simulator (1/3)
`S05`	`5 Oct`	Processes and Threads	Architecture simulator (2/3)
`S06`	`14 Oct` (1)	Processes and Threads	Architecture simulator (3/3)
`S07`	`19 Oct`	Processes and Threads	Development environment

(1) Wed, substitutes 12 Oct

Full schedule (II)

Session	Date	Theory (`NA8`)	Lab (`EL4`)
`S08`	`26 Oct`	Process Synchronization (1/3)	Introduction to Git
`S09`	`5 Nov` (2)	Process Synchronization (2/3)	Practice 1: explanation
`S10`	`9 Nov`	Process Synchronization (3/3)	**PEI#1**
`S11`	`16 Nov`	CPU Scheduling (1/3)	Device 2: explanation
`S12`	`23 Nov`	CPU Scheduling (2/3)	Development
`S13`	`30 Nov`	CPU Scheduling (3/3)	Development
`---`	`7 Dec`	Holiday (Inmaculada) — no session	Holiday
`S15`	`14 Dec`	Exercises	**PEI#2**

(2) Thu, substitutes 2 Nov

Exams and Grading

Grading

Continuous evaluation — 25/26 figures, to be confirmed in the 26/27 guía docente

- **Theory evaluation activities** `[25%]` — multiple choice tests, one per part
- **Laboratory practical activities** `[40%]` — architecture simulator, Linux console, programming environment
- **Problem resolution** `[35%]` — written exam with two practical problems

Exams

- **PEI#1** — Monday **9 November 2026**, 12:00–14:00, `EL4`
  - Theory 1→2 + Lab 1→4
- **PEI#2** — Monday **14 December 2026**, 12:00–14:00, `EL4`
  - Theory 3→5 + Lab 5

Academic integrity & AI

- **Plagiarism and cheating in exams** are **strictly forbidden** and academically sanctioned.
- **Generative AI** is **allowed and encouraged** in coursework, but its use **may be regulated or restricted at specific moments** (e.g. during exams/PEI) — this will always be stated explicitly.

Bibliography

- Sebastián Sánchez Prieto, **Sistemas Operativos**, Servicio de publicaciones de la Universidad de Alcalá, 2005.
- Abraham Silberschatz, Peter B. Galvin, Greg Gagne, **Operating System Concepts**, 10th Edition, Wiley, 2018.
- Andrew S. Tanenbaum, Herbert Bos, **Modern Operating Systems**, 4th Edition, Pearson, 2014.
- William Stallings, **Operating Systems: Internals and Design Principles**, 9th Edition, Pearson, 2018.

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Questions?